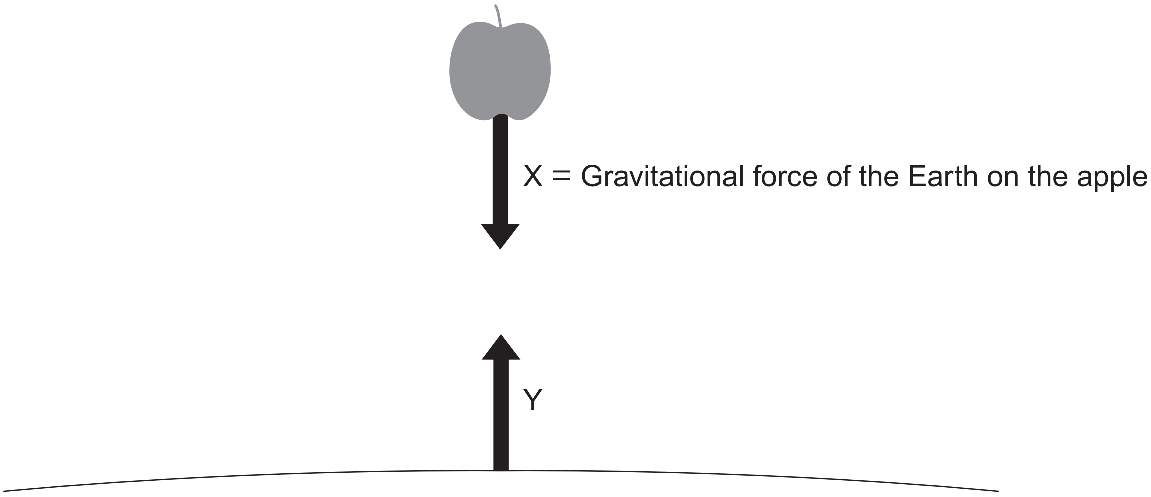
Examiner only

3. Newton's third law states that if a body A exerts a force on body B, body B exerts an equal and opposite force on body A.

(a) The diagram (not to scale) shows an apple of weight 1.5 N falling to the Earth.



(i) Underline the correct word in the brackets to describe force Y. [2]

Force Y is the (**gravitational / magnetic / mass**) force of the apple on the (**air / Earth / Moon**).

(ii) State how the two forces, X and Y, compare. [1]

....

(iii) Use an equation from page 2 to calculate the mass of the apple. [2]
(g = 10 N/kg = 10 m/s^2)

mass = ... kg

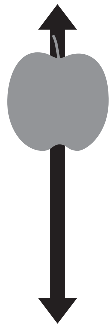
(b) The apple is dropped from rest.

(i) State the initial acceleration of the apple. [1]

acceleration = ... m/s^2

Examiner only

(ii) At some time later in the fall, the air resistance acting on the apple is 0.25 N.



Determine the size of the resultant force acting on the apple. [1]

resultant force = ... N

(iii) Use the equation:

resultant force = mass × acceleration

to calculate the new acceleration of the apple. [3]

acceleration = ... m/s^2

(c) The Apollo 15 astronaut David Scott performed an experiment on the Moon, where there is no air.
He predicted that if a hammer and a feather were dropped together from the same height, they would hit the Moon's surface at the same time.
Explain whether you agree with this prediction. [2]

....

....

....

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Turn over.